
Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere

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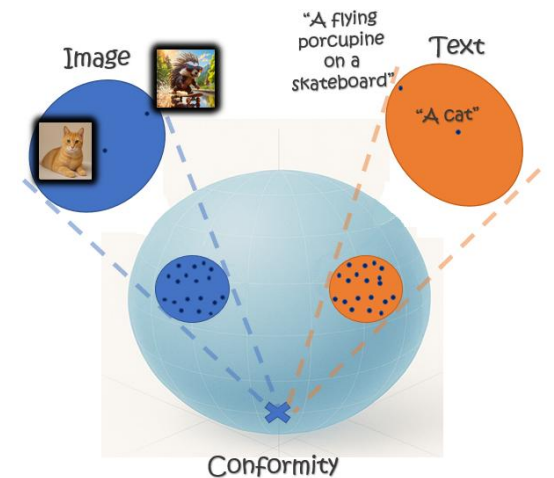
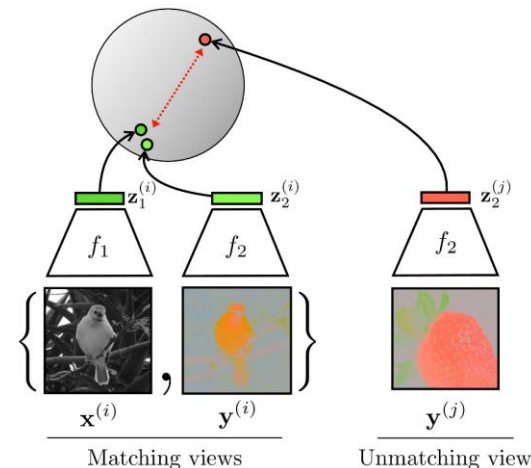
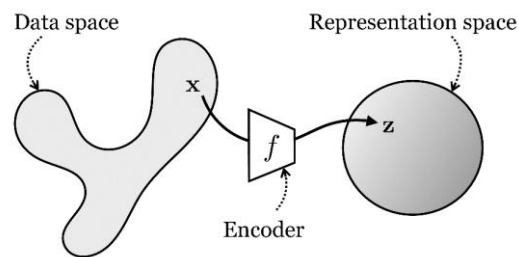
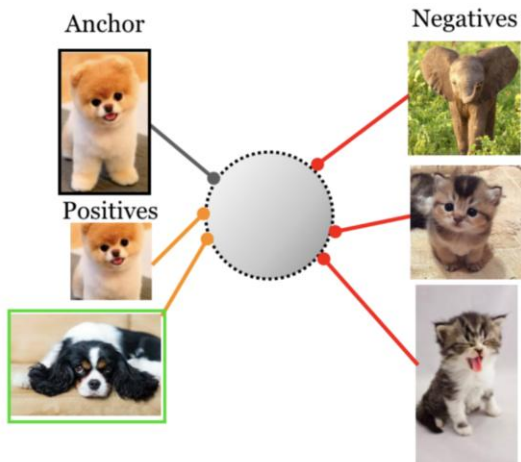
Abstract

Contrastive representation learning has been outstandingly successful in practice. In this work, we identify two key properties related to the contrastive loss: (1) alignment (closeness) of features from positive pairs, and (2) uniformity of the induced distribution of the (normalized) features on the hypersphere. We prove that, asymptotically, the contrastive loss optimizes these properties, and analyze their positive effects on downstream tasks. Empirically, we introduce an optimizable metric to quantify each property. Extensive experiments on standard vision and language datasets confirm the strong agreement between both metrics and downstream task performance. Directly optimizing for these two metrics leads to representations with comparable or better performance at downstream tasks than contrastive learning.

- Introduction
- Related Work
- Preliminaries on Unsupervised Contrastive Representation Learning
- Feature Distribution on the Hypersphere
- Experiments
- Discussion

Before review ... why I choose this paper?

1. I wondered why most papers draw the latent space as a hypersphere and what is the benefit of hypersphere-shaped latent space.
2. Contrastive loss is useful tools to obtain good representation. and this paper explains well how Contrastive loss works.
3. Nowadays most of multi modal task using CLIP(Contrastive Language Image Pre-trained)-like frameworks. The understanding of Contrastive learning is necessary if you want to engage multi modal tasks.



Before review ... There is a lot of math in this paper

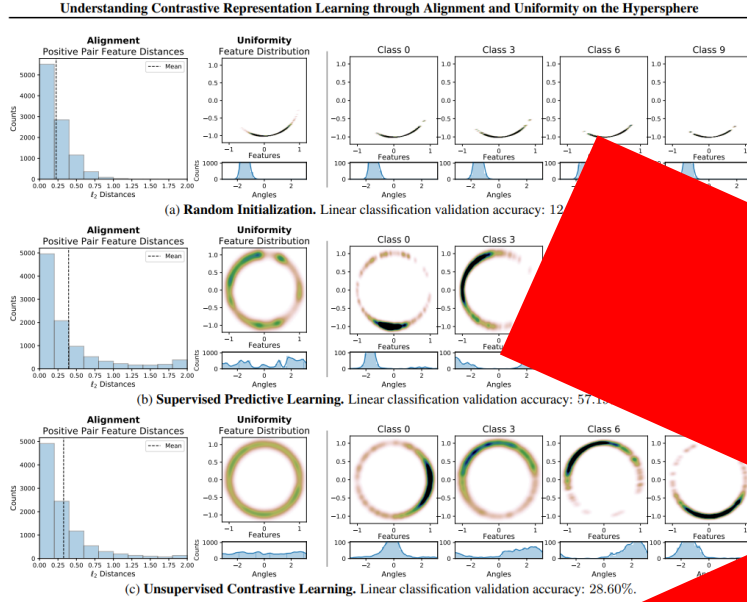


Figure 3: Representations of CIFAR-10 validation set on S^1 . **Alignment analysis:** We show distances between features of positive pairs (two random augmentations). **Uniformity analysis:** We show Gaussian kernel density estimation (KDE) in $\mathbb{R}^2$ and von Mises-Fisher (vMF) KDE on S^1 . **Four rightmost plots** visualize feature distributions of selected classes. Contrastive learning is both *aligned* (having low positive pair feature distances) and

of p , we can derive

$$\mathcal{L}_{\text{contrastive}}(f; \tau, M) = \mathbb{E}_{(x, y) \sim p_{\text{pos}}} [-f(x)^T f(y) / \tau] + \mathbb{E}_{\substack{(x, y) \sim p_{\text{data}} \\ \{x_i\}_{i=1}^M \sim p_{\text{data}}}} \left[\log \left(e^{f(x)^T f(y) / \tau} + \sum_i e^{f(x_i)^T f(y) / \tau} \right) \right].$$

Because the $\sum_i e^{f(x_i)^T f(y) / \tau}$ term is always positive and bounded below, the loss favors smaller $\mathbb{E} [-f(x)^T f(y) / \tau]$, i.e., having more aligned positive pair features. Suppose the encoder is perfectly aligned, i.e., $\mathbb{P}[f(x) = f(y)] = 1$, then minimizing the loss is equivalent to optimizing

$$\mathbb{E}_{\substack{x \sim p_{\text{data}} \\ \{x_i\}_{i=1}^M \sim p_{\text{data}}}} \left[\log \left(e^{1/\tau} + \sum_i e^{f(x_i)^T f(x) / \tau} \right) \right],$$

which is akin to maximizing the LogSumExp transformation of the pairwise potential. The LogSumExp transformation pushes features away from each other, thus encouraging roughly uniformly distributed features on S^d .

4.1. Quantifying Alignment

For further analysis, we need a measure alignment and uniformity. We propose the following two metrics (losses).

4.1.1. ALIGNMENT

The alignment loss is straightforwardly defined with the expected distance between positive pairs:

$$\mathcal{L}_{\text{align}}(f; \alpha) \triangleq \mathbb{E}_{(x, y) \sim p_{\text{pos}}} [\|f(x) - f(y)\|_2^\alpha], \quad \alpha > 0.$$

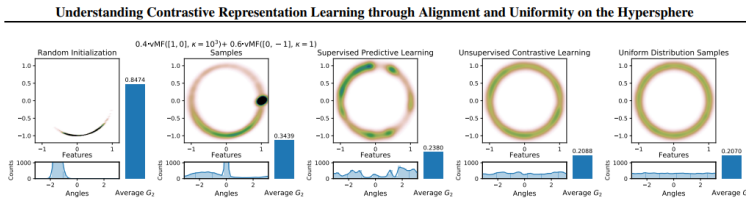


Figure 4: Average pairwise G_2 potential as a measure of uniformity. Each plot shows 10000 points distributed on S^1 obtained via either applying an encoder on CIFAR-10 validation set (same as those in Figure 3) or sampling from the uniform distribution on S^1 , as described in plot titles. We show the points with Gaussian KDE and the angles with vMF KDE.

Both asymptotically, the metric should be non-increasing as the number of samples goes to infinity.

There is a lot of math in this paper, but today: less math, more intuition.

For further analysis, we need a measure alignment and uniformity. We propose the following two metrics (losses).

The alignment loss is straightforwardly defined with the expected distance between positive pairs:

$$\min_{\mu \in \mathcal{M}(S^d)} \int \int G_t(u, v) d\mu.$$

Proof. See appendix. $\square$

In addition, as number of points goes to infinity, distributions of points minimizing the average pairwise potential converge weak* to the uniform distribution. Recall the definition of the weak* convergence of measures.

Definition (Weak* convergence of measures). A sequence of Borel measures $\{\mu_n\}_{n=1}^\infty$ in $\mathbb{R}^p$ converges weak* to a

it is clear that the uniform distribution is the unique minimizer of the expected pairwise potential. Theorem 1. For $\mathcal{M}(S^d)$ the set of Borel probability measures on S^d , σ_d is the unique solution of

Empirically, we evaluate the average pairwise potential on various finite point collections on S^1 in Figure 4. The values nicely align with our intuitive understanding of uniformity.

We further discuss properties of $\mathcal{L}_{\text{uniform}}$ and characterize its optimal value and range in the appendix.

4.2. Limiting Behavior of Contrastive Learning

In this section, we formalize the intuition that contrastive learning optimizes alignment and uniformity, and characterize its asymptotic behavior. We consider optimization problems over all measurable encoder functions from the p_{data} measure in $\mathbb{R}^n$ to the Borel space S^{m-1} .

Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere

We first define the notion of optimal encoders for each of these two metrics.

Definition (Perfect Alignment). We say an encoder f is perfectly aligned if $f(x) = f(y)$ a.s. over $(x, y) \sim p_{\text{pos}}$.

Definition (Perfect Uniformity). We say an encoder f is perfectly uniform if the distribution of $f(x)$ for $x \sim p_{\text{data}}$ is the uniform distribution on S^{m-1} .

Definition (Perfect Uniformity). We note that it is not enough to have perfect alignment, e.g., when the distribution of $f(x)$ is concentrated on a lower-dimensional than the feature space. We say an encoder f is perfectly uniform if p_{data} and p_{pos} are samples from a finite von Mises-Fisher (vMF) kernel density estimation (KDE):

Asymptotically, the metric should be non-increasing as the number of samples goes to infinity. Empirical work has established that larger number of negative samples consistently leads to better downstream task performances (Wu et al., 2018; Tian et al., 2019; He et al., 2019; Chen et al., 2020a), and often uses very large values (e.g., $M = 65536$ in He et al. (2019)). The following theorem nicely confirms that optimizing w.r.t. the limiting loss indeed requires both alignment and uniformity.

Theorem 1 (Asymptotics of $\mathcal{L}_{\text{contrastive}}$). For fixed $\tau > 0$, the number of negative samples $M \rightarrow \infty$, the (normalized) loss converges to

$$-\log M = \lim_{M \rightarrow \infty} \frac{1}{M} \log \left(e^{f(x)^T f(y) / \tau} + \sum_i e^{f(x_i)^T f(y) / \tau} \right) = -\log M.$$

Proof. See appendix. $\square$

For the case of $M = 1$, the loss converges to $-\log M = 0$.

Proof. See appendix. $\square$

Relation with $\mathcal{L}_{\text{uniform}}$. The proof of Theorem 1 in the appendix connects the asymptotic $\mathcal{L}_{\text{contrastive}}$ form with minimizing average pairwise Gaussian potential, i.e., minimizing $\mathcal{L}_{\text{uniform}}$. Compared with the second term of Equation (2), $\mathcal{L}_{\text{uniform}}$ essentially pushes the log outside the outer expectation, without changing the minimizer (perfectly uniform

encoders). However, due to its pairwise nature, $\mathcal{L}_{\text{uniform}}$ is much simpler in form and avoids the computationally expensive softmax operation in $\mathcal{L}_{\text{contrastive}}$ (Goodman, 2001; Bengio et al.; Gutmann & Hyvärinen, 2010; Grave et al., 2017; Chen et al., 2018).

Relation with feature distribution entropy estimation.

When p_{data} is uniform over finite samples $\{x_1, x_2, \dots, x_N\}$ (e.g., a collected dataset), the second term in Equation (2) can be alternatively viewed as a resubstitution entropy estimator of $f(x)$ (Ahmad & Lin, 1976), where x follows the underlying distribution p_{nature} that generates $\{x_i\}_{i=1}^N$ via a von Mises-Fisher (vMF) kernel density estimation (KDE):

$$\begin{aligned} \mathbb{E}_{x \sim p_{\text{data}}} \left[\log \mathbb{E}_{x' \sim p_{\text{data}}} \left[e^{f(x')^T f(x) / \tau} \right] \right] &= \frac{1}{N} \sum_{i=1}^N \log \left(\frac{1}{N} \sum_{j=1}^N e^{f(x_i)^T f(x_j) / \tau} \right) \\ &= \frac{1}{N} \sum_{i=1}^N \log \hat{p}_{\text{vMF-KDE}}(f(x_i)) + \log Z_{\text{vMF}} \\ &\triangleq -\hat{H}(f(x)) + \log Z_{\text{vMF}}, \quad x \sim p_{\text{nature}} \\ &\triangleq -\hat{I}(x; f(x)) + \log Z_{\text{vMF}}, \quad x \sim p_{\text{nature}}, \end{aligned}$$

where

- $\hat{p}_{\text{vMF-KDE}}$ is the KDE based on samples $\{f(x_i)\}_{i=1}^N$ using a vMF kernel with $\kappa = \tau^{-1}$,
- Z_{vMF} is the normalization constant for vMF distribution with $\kappa = \tau^{-1}$,
- $\hat{H}$ denotes the resubstitution entropy estimator,
- $\hat{I}$ denotes the mutual information estimator based on $\hat{H}$, since f is a deterministic function.

Relation with the InfoMax principle. Many empirical works are motivated by the InfoMax principle, i.e., maximizing $I(f(x); f(y))$ for $(x, y) \sim p_{\text{pos}}$. However, the interpretation of $\mathcal{L}_{\text{contrastive}}$ as a lower bound of $I(f(x); f(y))$ is known to be inconsistent with its actual behavior in practice (Tschannen et al., 2019). Our results instead analyze the properties of $\mathcal{L}_{\text{contrastive}}$ itself. Considering the identity $I(f(x); f(y)) = H(f(x)) - H(f(x) | f(y))$, we can see that while uniformity indeed favors large $H(f(x))$, alignment is stronger than merely desiring small $H(f(x) | f(y))$. In particular, both Theorem 1 and the above connection with maximizing an entropy estimator provide alternative interpretations and motivations that $\mathcal{L}_{\text{contrastive}}$ optimizes for aligned and information-preserving encoders.

Finally, even for the case where only a single negative sample is used (i.e., $M = 1$), we can still prove a weaker result, which we describe in details in the appendix.

- A vast number of recent empirical works learn representations with a unit l_2 norm constraint, effectively restricting the output space to the unit hypersphere, including many unsupervised contrastive representation learning methods.
- Intuitively, having the features live on the unit hypersphere leads to several desirable traits.
 - **Fixed-norm vectors** are known to **improve training stability** in modern machine learning where **dot products** are ubiquitous.
 - Moreover, if features of a class are sufficiently well clustered, they are **linearly separable** with the rest of feature space, a common criterion used to evaluate representation quality.

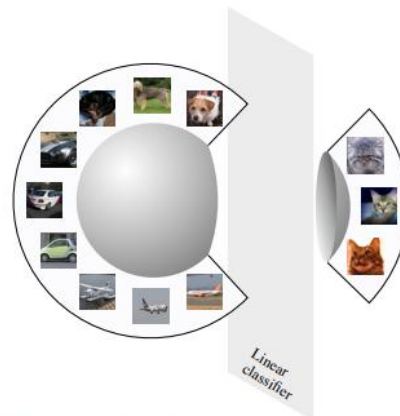


Figure 2: **Hypersphere**: When classes are well-clustered (forming spherical caps), they are linearly separable. The same does not hold for Euclidean spaces.

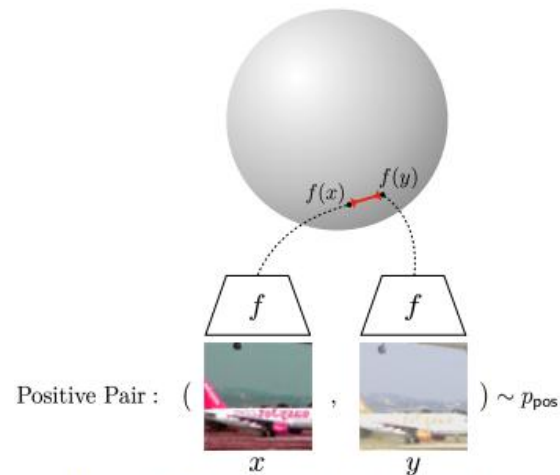
- **Alignment**

- Alignment favors encoders that assign similar features to similar samples.

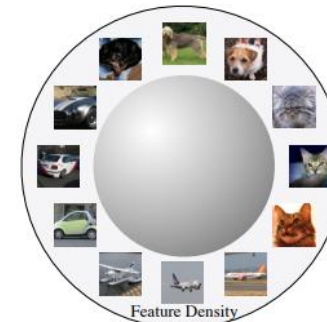
- **Uniformity**

- Uniformity prefers a feature distribution that preserves maximal information, i.e., the uniform distribution on the unit hypersphere.

- In this work, we analyze the **alignment** and **uniformity** properties.



Alignment: Similar samples have similar features.
(Figure inspired by Tian et al. (2019).)



Uniformity: Preserve maximal information.

Figure 1: Illustration of alignment and uniformity of feature distributions on the output unit hypersphere. STL-10 (Coates et al., 2011) images are used for demonstration.

- Unsupervised Contrastive Representation Learning(UCRL)
 - Previous: InfoMax principle (MI maximization)
 - Problem: Tighter MI bound $\rightarrow$ worse performance
 - What contrastive loss exactly does = mystery
 - Explains via geometric properties (alignment + uniformity)
- Representation learning on the unit hypersphere
 - Empirical success: VAE, face embedding
 - Known: Uniformly distributed points $\rightarrow$ good representations

- The popular unsupervised contrastive representation learning method learns representations from unlabeled data.

- Assumptions

- Symmetry: $\forall x, y, p_{pos(x,y)} = p_{pos(y,x)}$
- Matching marginal: $\forall x, \int p_{pos}(x, y) dy = p_{data}(x)$
- $f : R^n \rightarrow S^{m-1}$

- $$L_{contrastive} = -\mathbb{E} \left[\log \left(\frac{\exp\left(\frac{f(x)^T f(y)}{\tau}\right)}{\left(\exp\left(\frac{f(x)^T f(y)}{\tau}\right) + \sum_{\{i=1\}}^M \exp\left(\frac{f(x_i^-)^T f(y)}{\tau}\right)\right)} \right) \right]$$

$$\mathcal{L}_{contrastive}(f; \tau, M) \triangleq$$

$$\mathbb{E}_{\substack{(x,y) \sim p_{pos} \\ \{x_i^-\}_{i=1}^M \stackrel{\text{i.i.d.}}{\sim} p_{data}}} \left[-\log \frac{e^{f(x)^T f(y)/\tau}}{e^{f(x)^T f(y)/\tau} + \sum_i e^{f(x_i^-)^T f(y)/\tau}} \right],$$

(1)

- Necessity of normalization
 - Without the norm constraint, the softmax distribution can be made arbitrarily sharp by simply scaling all the features.
- The InfoMax principle
 - Many empirical works are motivated by the InfoMax principle of maximizing $I(f(x); f(y))$
 - However, maximizing a tighter bound often leads to worse downstream task performance

- **Alignment:** two samples forming a positive pair should be mapped to nearby features, and thus be (mostly) invariant to unneeded noise factors.
- **Uniformity:** feature vectors should be **roughly uniformly distributed** on the unit hypersphere S^{m-1} , preserving as much information of the data as possible.
- Experiments set
 - Dataset: CIFAR-10
 - Model: AlexNet / S^1 ($m = 2$)
 - Random Init / Supervised (Cross entropy) / Unsupervised (contrastive learning)
 - $\tau = 0.5$ / $M = 256$
 - standard data augmentations

Feature Distribution on the Hypersphere

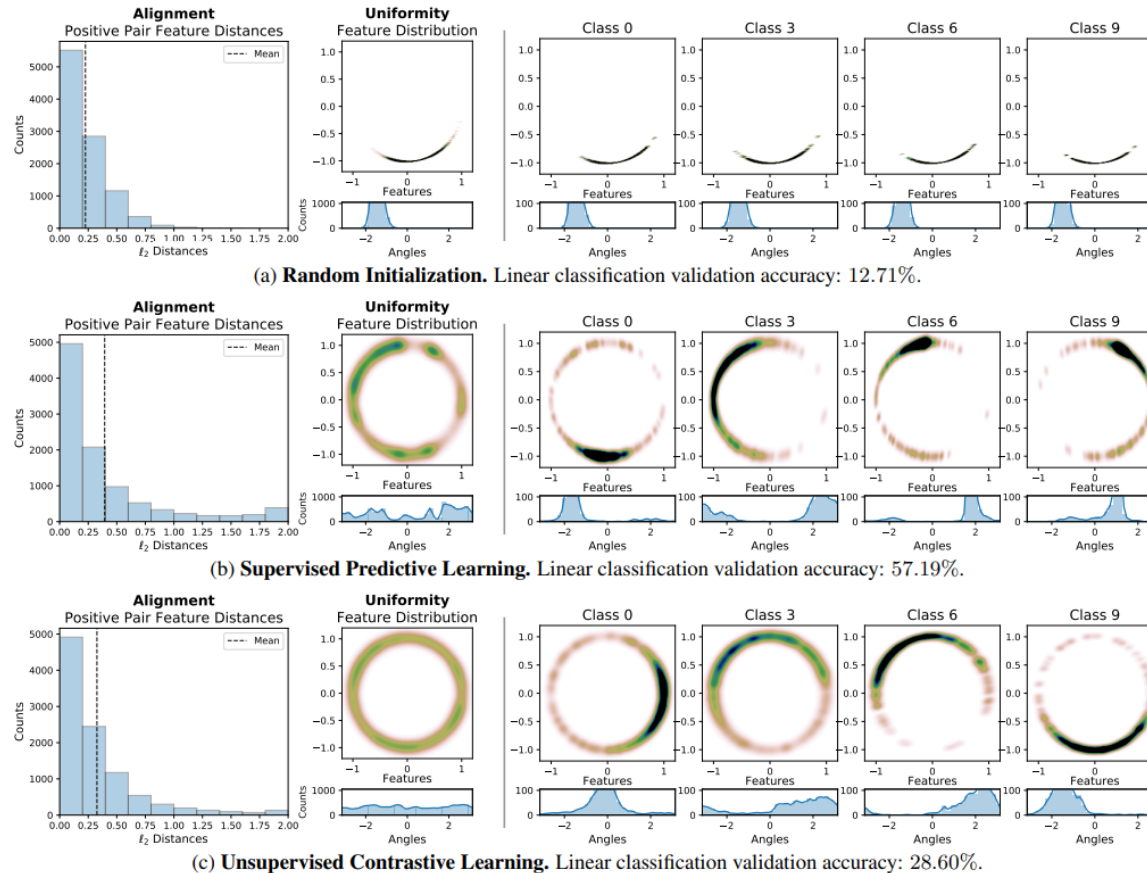


Figure 3: Representations of CIFAR-10 validation set on S^1 . **Alignment analysis:** We show distribution of distance between features of positive pairs (two random augmentations). **Uniformity analysis:** We plot feature distributions with Gaussian kernel density estimation (KDE) in $\mathbb{R}^2$ and von Mises-Fisher (vMF) KDE on angles (i.e., $\arctan2(y, x)$) for each point $(x, y) \in S^1$. **Four rightmost plots** visualize feature distributions of selected specific classes. Representation from contrastive learning is both *aligned* (having low positive pair feature distances) and *uniform* (evenly distributed on S^1).

$$\mathcal{L}_{\text{contrastive}}(f; \tau, M) = \mathbb{E}_{(x,y) \sim p_{\text{pos}}} [-f(x)^{\top} f(y) / \tau] \\ + \mathbb{E}_{\substack{(x,y) \sim p_{\text{pos}} \\ \{x_i^{-}\}_{i=1}^M \sim p_{\text{data}}}} \left[\log \left(e^{f(x)^{\top} f(y) / \tau} + \sum_i e^{f(x_i^{-})^{\top} f(x) / \tau} \right) \right]$$

Because the $\sum_i e^{f(x_i^{-})^{\top} f(x) / \tau}$ term is always positive and bounded below, the loss favors smaller $\mathbb{E} [-f(x)^{\top} f(y) / \tau]$, i.e., having more aligned positive pair features. Suppose the encoder is perfectly aligned, i.e., $\mathbb{P}[f(x) = f(y)] = 1$, then minimizing the loss is equivalent to optimizing

$$\mathbb{E}_{\substack{x \sim p_{\text{data}} \\ \{x_i^{-}\}_{i=1}^M \sim p_{\text{data}}}} \left[\log \left(e^{1/\tau} + \sum_i e^{f(x_i^{-})^{\top} f(x) / \tau} \right) \right],$$

$$\mathcal{L}_{\text{align}}(f; \alpha) \triangleq \mathbb{E}_{(x,y) \sim p_{\text{pos}}} [\|f(x) - f(y)\|_2^{\alpha}], \quad \alpha > 0.$$

$$\mathcal{L}_{\text{uniform}}(f; t) \triangleq \log \mathbb{E}_{x,y \stackrel{\text{i.i.d.}}{\sim} p_{\text{data}}} [G_t(u, v)] \\ = \log \mathbb{E}_{x,y \stackrel{\text{i.i.d.}}{\sim} p_{\text{data}}} \left[e^{-t \|f(x) - f(y)\|_2^2} \right], \quad t > 0.$$

```
# bsz : batch size (number of positive pairs)
# d   : latent dim
# x   : Tensor, shape=[bsz, d]
#     : latents for one side of positive pairs
# y   : Tensor, shape=[bsz, d]
#     : latents for the other side of positive pairs
# lam : hyperparameter balancing the two losses

def lalign(x, y, alpha=2):
    return (x - y).norm(dim=1).pow(alpha).mean()

def lunif(x, t=2):
    sq_pdist = torch.pdist(x, p=2).pow(2)
    return sq_pdist.mul(-t).exp().mean().log()

loss = lalign(x, y) + lam * (lunif(x) + lunif(y)) / 2
```

Figure 6: PyTorch implementation of $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{uniform}}$.

- Datasets
 - STL-10, NYU-Depth-V2, ImageNet / ImageNet-100, BookCorpus
- Model
 - CNN, RNN, MoCo, Quick-Thought Vectors
- $L_{contrastive}$, L_{align} , and/or $L_{uniform}$, with varying
 - (possibly zero) weights on the three losses,
 - temperature τ for $L_{contrastive}$,
 - $\alpha \in \{1, 2\}$ for L_{align} ,
 - $t \in \{1, 2, \dots, 8\}$ for $L_{uniform}$,
 - batch size (affecting the number of (negative) pairs for $L_{contrastive}$ and $L_{uniform}$),
 - embedding dimension,
 - number of training epochs and learning rate,
 - initialization (from scratch vs. a pretrained encoder).

$$\mathcal{L}_{align}(f; \alpha) \triangleq \mathbb{E}_{(x,y) \sim p_{pos}} [\|f(x) - f(y)\|_2^\alpha]$$
$$\mathcal{L}_{uniform}(f; t) \triangleq \log \mathbb{E}_{x,y \stackrel{i.i.d.}{\sim} p_{data}} \left[e^{-t\|f(x)-f(y)\|_2^2} \right].$$

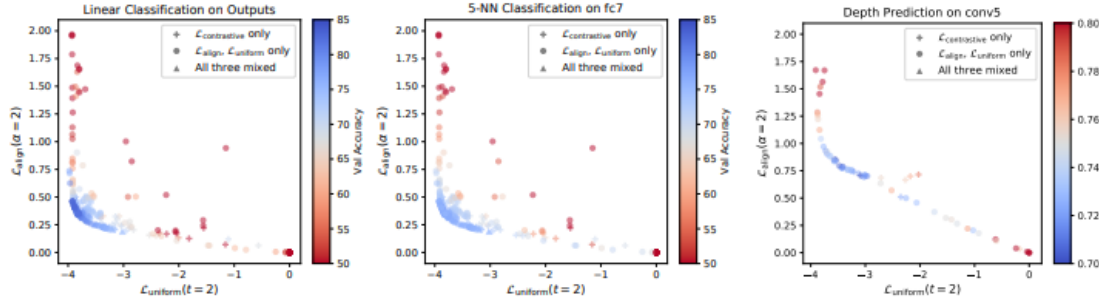


Figure 5: Metrics and performance of STL-10 and NYU-DEPTH-V2 experiments. Each point represents a trained encoder, with its x - and y -coordinates showing $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{uniform}}$ metrics and color showing the performance on validation set. **Blue** is better for both tasks. Encoders with low $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{uniform}}$ are consistently the better performing ones (lower left corners).

	Loss Formula	Validation Set Accuracy $\uparrow$			
		Output + Linear	Output + 5-NN	fc7 + Linear	fc7 + 5-NN
Best $\mathcal{L}_{\text{contrastive}}$ only	$\mathcal{L}_{\text{contrastive}}(\tau=0.19)$	80.46%	78.75%	83.89%	76.33%
Best $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{uniform}}$ only	$0.98 \cdot \mathcal{L}_{\text{align}}(\alpha=2) + 0.96 \cdot \mathcal{L}_{\text{uniform}}(t=2)$	81.15%	78.89%	84.43%	76.78%
Best among all encoders	$\mathcal{L}_{\text{contrastive}}(\tau=0.5) + \mathcal{L}_{\text{uniform}}(t=2)$	81.06%	79.05%	84.14%	76.48%

Table 1: STL-10 encoder evaluations. Numbers show linear and 5-nearest neighbor (5-NN) classification accuracies on the validation set. The best result is picked by encoder outputs linear classifier accuracy from a 5-fold training set cross validation, among all 150 encoders trained from scratch with 128-dimensional output and 768 batch size.

	Loss Formula	Validation Set MSE $\downarrow$	
		conv5	conv4
Best $\mathcal{L}_{\text{contrastive}}$ only	$0.5 \cdot \mathcal{L}_{\text{contrastive}}(\tau=0.1)$	0.7024	0.7575
Best $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{uniform}}$ only	$0.75 \cdot \mathcal{L}_{\text{align}}(\alpha=2) + 0.5 \cdot \mathcal{L}_{\text{uniform}}(t=2)$	0.7014	0.7592
Best among all encoders	$0.75 \cdot \mathcal{L}_{\text{align}}(\alpha=2) + 0.5 \cdot \mathcal{L}_{\text{uniform}}(t=2)$	0.7014	0.7592

Table 2: NYU-DEPTH-V2 encoder evaluations. Numbers show depth prediction mean squared error (MSE) on the validation set. The best result is picked based on conv5 layer MSE from a 5-fold training set cross validation, among all 64 encoders trained from scratch with 128-dimensional output and 128 batch size.

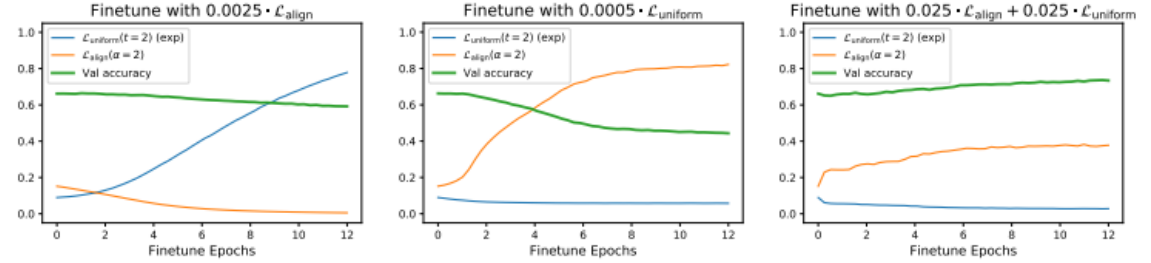
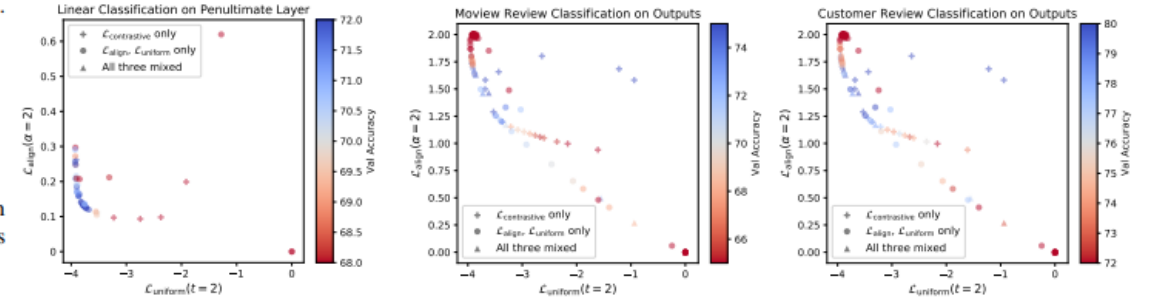


Figure 8: Finetuning trajectories from a STL-10 encoder trained with $\mathcal{L}_{\text{contrastive}}$ using a suboptimal temperature $\tau = 2.5$. Finetuning objectives are weighted combinations of $\mathcal{L}_{\text{align}}(\alpha=2)$ and $\mathcal{L}_{\text{uniform}}(t=2)$. For each intermediate checkpoint, we measure $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{uniform}}$ metrics, as well as validation accuracy of a linear classifier trained from scratch on the encoder outputs. $\mathcal{L}_{\text{uniform}}$ is exponentiated for plotting purpose. **Left and middle:** Performance degrades if only one of alignment and uniformity is optimized. **Right:** Performance improves when both are optimized.



(a) 45 IMAGENET-100 encoders are trained with MoCo-based methods, and evaluated with linear classification. (b) 108 BOOKCORPUS encoders are trained with Quick-Thought-Vectors-based methods, and evaluated with logistic binary classification on Movie Review Sentence Polarity and Customer Product Review Sentiment tasks.

Figure 9: Metrics and performance of IMAGENET-100 and BOOKCORPUS experiments. Each point represents a trained encoder, with its x - and y -coordinates showing $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{uniform}}$ metrics and color showing the validation accuracy. **Blue** is better. Encoders with low $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{uniform}}$ consistently perform well (lower left corners), even though the training methods (based on MoCo and Quick-Thought Vectors) are different from directly optimizing the contrastive loss in Equation (1).

- Through theoretical analysis and extensive experiments
 - contrastive loss with the alignment and uniformity properties are relative.
 - and confirm their strong connection with downstream task performances.
 - directly optimizing proposed metrics often leads to representations of better quality.
- Niceness of the unit hypersphere
 - the question of why the unit hypersphere is a nice feature space is not answered.
 - One possible direction is to formalize the intuition that connected sets with smooth boundaries are nearly linearly separable in the hyperspherical geometry.
 - linear separability is one of the most widely used criteria for representation quality.

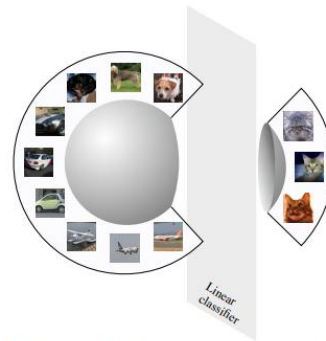


Figure 2: **Hypersphere:** When classes are well-clustered (forming spherical caps), they are linearly separable. The same does not hold for Euclidean spaces.